Roomify

Deployment

Proposed hosting, cost and steps (proposal only, not deployed)

Team Bootcamp2Roomify
Release candidate v0.9.0-rc1
24 October 2026
github.com/Bootcamp2Roomify/roomify_frontend · roomify_backend · roomify_cv

Deployment

Status: proposal only. No cloud resources have been created. Every step below needs the project owner's approval and their own credentials.

Recommendation: Vercel + Fly.io + Neon + existing S3

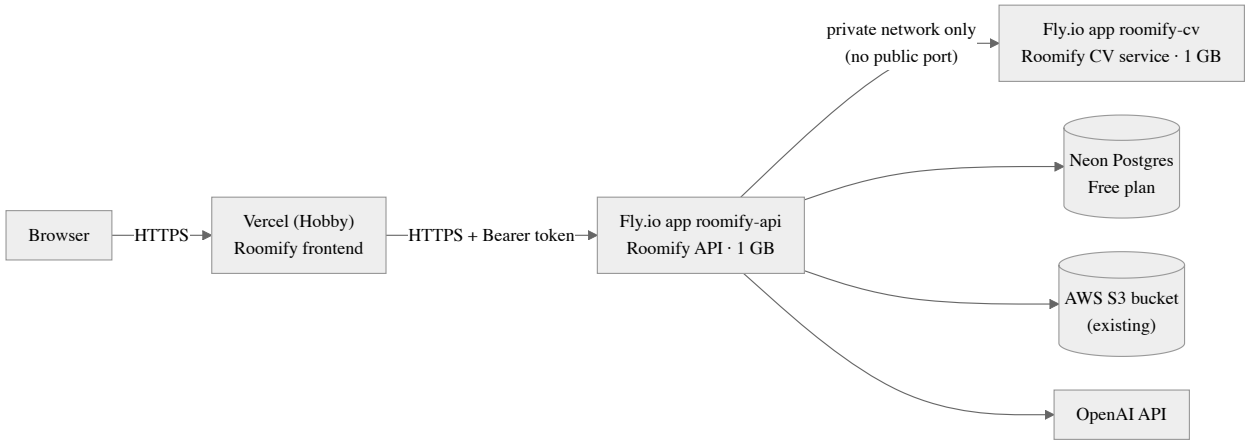


Figure 1. Proposed Roomify deployment: Vercel for the frontend, Fly.io for the API and a private CV service, Neon Postgres, the existing S3 bucket and the OpenAI API.

Why this shape:

- **Each service already ships as a container** (backend and CV Dockerfiles are pinned and tested), so the hosts only run images: no build-system work.
- **The CV service gets no public address.** It has no authentication and an `image_url` input; Fly's private network lets only the backend reach it.
- **Long AI calls fit.** Image edits can take up to ~2 minutes. Vercel Hobby functions with Fluid compute allow 300 s; the backend's AI timeout is 180 s. The browser never calls the backend directly (the Next bridge does), so no CORS changes are needed.
- **Measured memory fits 1 GB machines:** CV container 374 MiB after inference, backend 424 MiB after start-up (Docker, 2026-10-24).

Cost (list prices checked 2026-10-24)

Piece	Plan	Monthly
Frontend	Vercel Hobby (personal/non-commercial — fine for judging)	$0
Backend	Fly.io shared-cpu-1x, 1 GB, always on, `iad`	$6.70
CV service	Fly.io shared-cpu-1x, 1 GB, always on, `iad`	$6.70
Database	Neon Free (0.5 GB storage, scales to zero after 5 min idle)	$0
Images	Existing AWS S3 bucket (Standard ≈ $0.023/GB-month)	< $0.10
Infrastructure total		**≈ $13.40 / month**
OpenAI usage	per redesign: plan + concept + suggestions + DIY	**Pending: requires OpenAI key** (Evaluation Evidence §2)

Notes: Fly bills per second, so stopping the machines after judging stops the cost. Fly raised its RAM price on 2026-10-01 (1 GB machine $5.70 → $6.70); Frankfurt is $7.73. Neon's first query after idling has a cold start of up to a few seconds. Sources: Fly.io pricing, Vercel function limits, Neon pricing.

Steps (to run after approval)

Prerequisites the owner provides: Fly.io account (`fly auth login`), Vercel account linked to the GitHub org, Neon account, the existing S3 bucket name/region and an IAM key **scoped to that bucket**, an OpenAI key.

1. **Database** — Neon console → *New project* `roomify` (region AWS us-east-1) → copy host, database, user, password.
2. **CV service** (private):

```
cd roomify_cv
fly launch --name roomify-cv --region iad --no-deploy --internal-port 8000
# remove the [http_service] section from fly.toml so it has no public port
fly scale vm shared-cpu-1x --memory 1024 -a roomify-cv
fly deploy -a roomify-cv
```

3. **Backend**:

```
cd roomify_backend
fly launch --name roomify-api --region iad --no-deploy --internal-port 8080
fly secrets set -a roomify-api \
  DB_HOST=<neon-host> DB_PORT=5432 DB_NAME=<db> DB_USERNAME=<user> DB_PASSWORD=
<password> \
  JWT_SECRET=$(openssl rand -hex 32) OPENAI_API_KEY=<key> \
  STORAGE_TYPE=s3 S3_BUCKET=<bucket> AWS_REGION=<region> AWS_ACCESS_KEY_ID=<id>
AWS_SECRET_ACCESS_KEY=<secret> \
  CV_SERVICE_URL=http://roomify-cv.internal:8000
fly scale vm shared-cpu-1x --memory 1024 -a roomify-api
fly deploy -a roomify-api
curl https://roomify-api.fly.dev/api/health
```

Neon requires TLS; if the JDBC URL needs it explicitly, add `?sslmode=require` via a `SPRING_DATASOURCE_URL` secret.

4. **Frontend** — Vercel → *Add New Project* → import `Bootcamp2Roomify/roomify_frontend` → framework Next.js → environment variable `ROOMIFY_API_URL=https://roomify-api.fly.dev` → *Deploy*.
5. **Smoke test** — run the live browser suite against the deployment: `LIVE_BASE_URL=https://<project>.vercel.app LIVE_AI=1 npx playwright test -c playwright.live.config.ts`.
6. **After judging** — `fly scale count 0 -a roomify-api -a roomify-cv` (stops billing), or `fly apps destroy`.

Risks

- `gpt-4o-mini` (default text model) is shut down by OpenAI on **2027-01-14**; switch to `gpt-6-luna` with `AI_TEMPERATURE=` and `AI_REASONING_EFFORT=low` before then (config only).
- Vercel Hobby is for non-commercial use; move to Pro ($20/month) for anything beyond judging.
- A public demo can spend the OpenAI budget: set a monthly usage limit in the OpenAI dashboard.